

Chapter 3 — Control

Acting on what Formation described

Every technique in this chapter steers something that was rated. That is why it is here and not before Chapter 2.

Why this chapter comes second

- Technique taught before mechanism produces cargo-cult prompting: a list of tricks, no way to tell a good one from a superstition, and nothing to fall back on when one stops working.
- You have just spent Chapter 2 inside the rating process. You know what was rewarded.
- So each technique here comes with the same question: **what was rated, that this steers?**

And the ones that have no answer to that question get flagged, not taught.

Before you tune a single word

The vendor's own documentation states what it assumes you already have:

- A clear definition of the success criteria for your task
- Some way to test empirically against those criteria
- A first draft prompt to improve

Most prompting advice skips straight past the middle item. Without it you cannot tell an improvement from a coincidence — and neither can the person who told you the trick.

Thread opened — verification. Chapter 4 builds the missing middle item.

Be specific

The documentation's own test, and it is a good one:

Show your prompt to a colleague with minimal context on the task and ask them to follow it. If they'd be confused, Claude will be too.

- Be explicit about the output format and the constraints.
- Where order or completeness matters, give the steps as a numbered list.
- If you want effort beyond the obvious, ask for it. Do not expect it to be inferred.

Maps to: instruction following — the dimension your Chapter 2 rubric scored hardest. A vague prompt cannot be failed on instruction following, because there was no instruction to follow.

Say why, not just what

A constraint with a reason attached generalises. A bare rule does not.

Weaker

Always state units.

Stronger

Always state units. This goes into a report where a reader may convert the value, and a bare number gets converted wrongly.

- The second version covers cases you did not enumerate — a table, a figure caption, an appendix.
- The documentation puts it plainly: the model generalises from the explanation.

Structure the input

When a prompt mixes instructions, background, data and examples, tag each part so they cannot be confused with one another.

```
<instructions>
  Check the calculation below and report every step where a unit is dropped.
</instructions>

<context>
  This is from a draft methods section. It will be read by a reviewer.
</context>

<input>
  ... the working to be checked ...
</input>
```

- Use consistent, descriptive tag names. Nest them when the content has a real hierarchy.
- **This is also your first defence against prompt injection.** If instructions and data are visibly separated, text arriving inside `<input>` is easier to treat as data. Chapter 6 shows how far that gets you, which is not as far as you would like.

Examples do more than instructions

Examples are the most reliable way to steer format, tone and structure.

- **Relevant** — mirror the real task, not a toy version of it.
- **Diverse** — cover edge cases, and vary enough that no unintended pattern is picked up.
- **Structured** — wrap each in `<example>` tags so they are not mistaken for instructions.
- Three to five is the documented sweet spot.

Maps to: house style. In Chapter 2 you were handed a rewrite checklist and told to apply it without explanation. Examples are the same mechanism, pointed the other way — you are now the one specifying the voice.

Negative examples, and their cost

Showing what you do **not** want works, and it carries a risk the positive case does not.

- A negative example still puts the unwanted pattern into the context window.
- Prefer stating the desired behaviour. The documentation's rule: **tell it what to do instead of what not to do.**
- Where a negative example is genuinely clearer, label it unmistakably and pair it with the corrected version.

Role — and exactly what is being claimed

Setting a role in the system prompt is the most repeated advice in circulation.

- The documentation says a role **focuses behaviour and tone** for your use case, and that even one sentence makes a difference.
- Read that carefully. It is a claim about *behaviour and tone*.
- It is **not** a claim that a persona makes the model more factually correct.

"You are an expert structural engineer" — does that make the answer more **accurate**?

That is a different claim, it is testable, and **Chapter 4 tests it**. Do not settle it here from intuition, and notice how badly you want to.

Requesting reasoning — read the original scope

Chain-of-thought prompting is real and it is well evidenced. It is also routinely described as something it was not.

- The original result: prompting with **eight worked chain-of-thought exemplars**, on a 540-billion-parameter model, improving arithmetic, commonsense and symbolic reasoning.
- The paper states the ability emerges in *sufficiently large* models.
- That is a result about supplying worked examples. It is **not** a result about typing "*think step by step*" at a model that already reasons before answering.

Whether the folk version still buys you anything on the model you actually use is, again, **Chapter 4**.

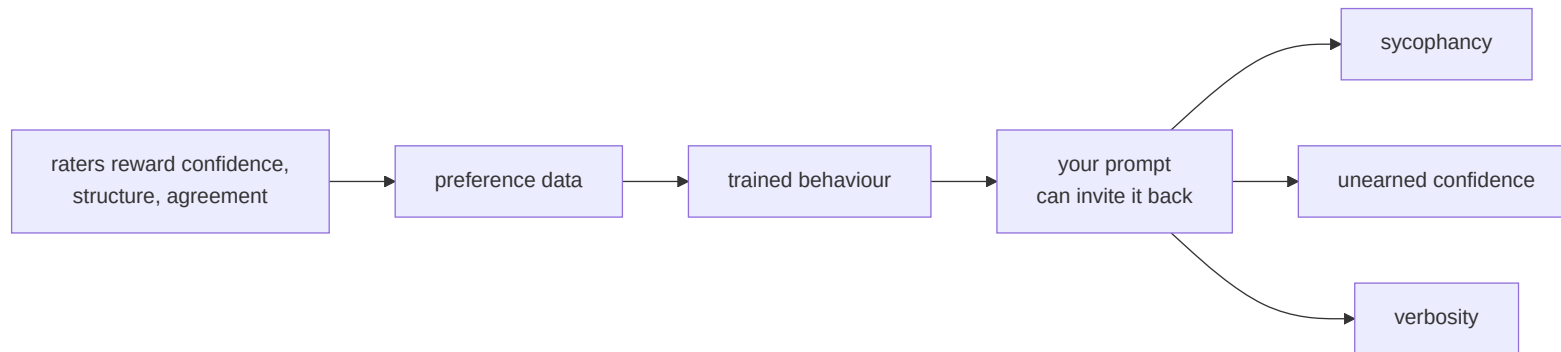
Thinking and effort

- Current models use **adaptive thinking**: the model decides when and how much to think, calibrated by an `effort` setting and by how hard your query is.
- Higher effort elicits more thinking. Easy queries get answered directly.
- More thinking is not free — it costs tokens and latency, which is Chapter 7’s material and Chapter 13’s bill.

Two version-specific facts, true as fetched and worth re-checking before you rely on them. `budget_tokens` is deprecated and returns an error on recent models — use `effort`, or `max_tokens` as a hard ceiling. And on Opus 5, raising or lowering effort does **not** reliably change visible response length; ask for concision explicitly instead.

You are on the other side of the rubric now

In Chapter 2 you rated. Under time pressure, you rewarded confidence, structure and length. So did everyone before you, at scale, and that became preference data.



The next three slides are not style advice. They are ways of accidentally asking for the behaviour you were warned about.

Anti-pattern: the leading question

Invites agreement

"I think this approach is the right one — do you agree?"

"Confirm that this method is appropriate here."

Invites analysis

"Give the strongest case against this approach, then the strongest case for it."

"Under what conditions would this method be the wrong choice?"

- Agreement was rewarded during rating. Asking a question with a preferred answer visible in it is asking for the trained behaviour.
- You will not notice, because the answer will agree with you.

Anti-pattern: asking for a verdict

- *"Is this correct?"* returns a judgement, and a judgement can be delivered confidently on no evidence.
- *"Work through this and show where it fails, or state that you found nothing"* returns something you can check.
- The second is longer, slower, and the only one of the two you can audit.

A verdict is cheap to produce and expensive to verify. An analysis is the other way round.

Forward reference — Chapter 5. The stated reasoning is not a guarantee of the actual mechanism. Asking for the working helps you catch errors; it does not prove the working caused the answer.

Anti-pattern: constraint stacking

"Summarise this in under 100 words. Include every assumption. Cite each source. Do not leave anything material out."

- These requirements conflict. Something has to give, and you did not say which.
- The model will not usually stop and tell you the brief is impossible. It will satisfy the ones it can and quietly drop the rest.
- **Name the priority explicitly:** which constraint wins when they collide.

This is the same failure as an over-specified rubric, seen from the author's side. In Chapter 2 an impossible rubric produced noisy ratings. Here it produces a confident answer that silently missed half the brief.

Prompts do not port

Everything on the preceding slides is calibrated to one provider's models.

- Tag conventions, role handling, thinking and effort controls, default verbosity — all provider-specific, and several are specific to a *version*.
- A prompt tuned on one model and moved to another is an untested prompt, not a working one.
- The documentation itself is organised model by model, and already carries exceptions for individual releases.

Sets up Chapter 12. When you reach outside this session — a second provider, a retrieval tool, an agent someone else wrote — you are re-testing, not reusing.

What you can take on trust, and what you cannot

	Standing
Be specific; state constraints; give the steps in order	Documented, and it maps to a rated dimension
Explain why a constraint exists	Documented, with a stated mechanism
Structure with tags; 3–5 relevant, diverse examples	Documented, with stated bounds
Tell it what to do rather than what not to do	Documented
A role improves tone and behaviour	Documented
A role improves accuracy	Not established. Chapter 4.
<i>"Think step by step"</i> helps a modern reasoning model	Not established. Chapter 4.

The bottom two rows are the reason the next chapter exists. Everything above them you can act on tomorrow; those two you should test before you believe.

Where this leaves us

- Technique, with a mechanism attached to each piece of it.
- Three anti-patterns that are not style errors — they are requests for trained behaviour you already watched being trained.
- Two widely repeated claims, deliberately left open.

You now have claims. Chapter 4 is how you test one.